

Floquet–Lyapunov consistency and matched sensing bounds for flux-controlled hyperchaos in dissipative optomechanics

Stella Rolande Mbokop Tchounda and Carolle Tchodimou,¹ Philippe Djourwe,² Sifeu Takougang Kingni,³ and Serge Guy Nana Engo¹

¹*Department of Physics, Faculty of Science, University of Yaoundé I, P.O. Box 812, Yaoundé, Cameroon*

²*Department of Physics, Faculty of Science, University of Ngaoundéré, P.O. Box 454, Ngaoundéré, Cameroon*

³*Department of Mechanical, Petroleum and Gas Engineering, National Advanced School of Mines and Petroleum Industries, University of Maroua, P.O. Box 46, Maroua, Cameroon*

Periodically driven optomechanical networks combine radiation-pressure nonlinearity, dissipation, and synthetic gauge phases in a single tunable platform. We establish a phase-consistent Floquet–Lyapunov protocol for a six-dimensional semiclassical model with one optical and two mechanical modes, and use it to separate numerical consistency from dynamical instability and measurement sensitivity. At a validated reference operating point, the analytic Jacobian agrees with finite differences to 3×10^{-10} , the one-period orbit residual is below 9×10^{-9} , and the maximum Floquet–Lyapunov discrepancy is 2.68×10^{-4} . A long-window grid of nine synthetic-flux phases with three initial-condition replicates (27 runs) satisfies all declared consistency criteria and shows no positive exponent: within the tested weak-coupling domain no flux-driven transition occurs. Strong coupling changes this picture. Classifying the complete six-exponent spectrum resolves a drive-gated transition from stable dynamics to chaos and then hyperchaos, entered through a Neimark–Sacker torus bifurcation of the drive-locked orbit, in which the synthetic-flux phase controls the number of simultaneously unstable directions (up to four); a three-seed check, a convergence probe, and a noise-resolved ensemble check (measured cavity quadrature 7.6–17 times the noise floor in the chaotic sector) support the classification at the tested points. A matched classical measurement-Fisher calculation for weak force sensing gives a maximum gain of 1.323 over the flux-off coupled reference and 1.159 over the single-mode reference: the enhancement is real but modest, and no claim of quantum advantage, SQL beating, Wigner negativity, or exceptional-point sensing is made. Both results are model-level demonstrations—the strong-coupling regime lies $2542\times$ beyond the anchored optomechanical coupling.

Keywords: optomechanics; synthetic gauge fields; Floquet dynamics; Lyapunov spectrum; hyperchaos; classical Fisher information

I. INTRODUCTION

Nonlinear optomechanical systems combine radiation-pressure coupling, coherent mechanical motion, and dissipation in a platform that can be driven far from stationarity^{1–6}. Periodic modulation adds a Floquet phase^{7–10}, while complex hopping amplitudes provide a route to synthetic gauge structure^{11,12}. These ingredients make it possible to ask a sharper question than whether a particular trajectory appears irregular: which invariant control parameter changes the attractor structure, and which signatures survive independent stability and noise checks?

Recent work has already established that synthetic magnetism can influence complex optomechanical dynamics^{13,14}, while fluctuations can qualitatively alter the visibility of chaotic motion¹⁵. Identifying chaos in such systems also demands more than a single irregular trace or spectrum: established practice combines stability, phase-space, Poincaré, bifurcation, and Lyapunov diagnostics^{4,16–23}, and multimode cavities can host several simultaneously unstable directions whose resolution requires the full spectrum rather than a single largest exponent^{24,25}. This work is therefore not framed

around the generic statement that a synthetic field “creates chaos”. Its contribution is testable and distinct from those phase-scan studies: (i) a phase-convention-aware protocol in which monodromy multipliers, the full six-exponent Lyapunov spectrum, and the dissipative volume balance certify one another on the same orbit; (ii) a full-spectrum classification of a drive- and coupling-gated transition to hyperchaos in which the synthetic-flux phase controls the number of simultaneously unstable directions, at the level required for multimode systems²⁴; and (iii) an operational assessment, with matched resources, of whether a flux-induced change remains visible in a noisy measurement record, which directly bounds the sensor-performance claims advanced for this model class¹⁴. The central comparison is between a weak-coupling stable sector and an explicitly strong-coupling sector in which the number of unstable directions, rather than only the largest exponent, is resolved. The distinction is essential because local trajectory separation is not, by itself, evidence for an increase in Fisher information or a quantum metrological advantage.

We address three questions. First, does the gauge-invariant loop phase provide a reproducible control coordinate for the driven nonlinear dynamics? Second, do

monodromy multipliers and long-time Lyapunov exponents agree when they describe the same periodic orbit and formulation? Third, after thermal, vacuum, and detector noise are included, does a declared measurement statistic retain a flux-dependent signature relative to a matched reference? These questions are narrower than a claim of universal chaos enhancement.

II. MODEL AND CONVENTIONS

We use the rotating-frame Hamiltonian

$$\frac{H}{\hbar} = -\Delta a^\dagger a + \sum_{j=1}^2 \omega_j b_j^\dagger b_j - \sum_{j=1}^2 g_j a^\dagger a (b_j + b_j^\dagger) + J \left(e^{i\theta} b_1^\dagger b_2 + e^{-i\theta} b_2^\dagger b_1 \right) + i(E a^\dagger - E^* a) \quad (1)$$

with one cavity mode a and two mechanical modes b_j . The detuning convention is $\Delta = \omega_c - \omega_L$. Normalized quadratures obey $[X, Y] = i$ and have vacuum variance $1/2$.

After the declared semiclassical factorisation, the complex amplitudes obey

$$\dot{\alpha} = \left(i\Delta - \frac{\kappa}{2} \right) \alpha + i \sum_j g_j (\beta_j + \beta_j^*) \alpha + E(t), \quad (2)$$

$$\dot{\beta}_1 = - \left(i\omega_1 + \frac{\gamma_1}{2} \right) \beta_1 + i g_1 |\alpha|^2 - i J e^{i\theta} \beta_2, \quad (3)$$

$$\dot{\beta}_2 = - \left(i\omega_2 + \frac{\gamma_2}{2} \right) \beta_2 + i g_2 |\alpha|^2 - i J e^{-i\theta} \beta_1. \quad (4)$$

The real state is $x = (\alpha_r, \alpha_i, \beta_{1r}, \beta_{1i}, \beta_{2r}, \beta_{2i})^T$. All time values are reported separately as physical time, normalized time, and drive-period count.

For the three-link loop, the phase coordinate used in the analysis is the orientation-declared invariant

$$\Phi_{\text{syn}} = \arg(g_1 J g_2^*) \pmod{2\pi}. \quad (5)$$

A redefinition of the mode phases may redistribute link phases but cannot change Φ_{syn} .

III. METHODS

A. Phase-augmented dynamics

For a periodic drive with phase ϕ , we write

$$\dot{x} = f(x, \phi; p), \quad \dot{\phi} = \Omega, \quad (6)$$

where p denotes the physical and normalized parameter set. This formulation is kept distinct from the six-dimensional explicitly forced flow and from the stroboscopic map. The Jacobian of the augmented flow is

$$J_{\text{aug}} = \begin{pmatrix} \partial_x f & \partial_\phi f \\ 0 & 0 \end{pmatrix}. \quad (7)$$

B. Model and implementation validation

The analytic Jacobian $J = \partial f / \partial x$ is compared column by column with central finite differences at multiple states and drive phases. The trace is computed from the implemented equations and compared with the expected dissipative balance

$$\text{Tr } J = -\kappa - \gamma_1 - \gamma_2, \quad (8)$$

when the declared damping convention is used. This is a validation consequence, not a hard-coded numerical result.

C. Periodic orbit and Floquet analysis

A drive-locked periodic orbit is obtained from the fixed point of the one-period Poincaré map. Around the converged orbit, the fundamental matrix satisfies

$$\dot{\Phi}(t) = A(t)\Phi(t), \quad \Phi(0) = I, \quad M(T) = \Phi(T), \quad (9)$$

where $A(t) = J[x_p(t), t]$. The Floquet multipliers are the eigenvalues of $M(T)$, with rates $\rho_i = T^{-1} \log |\mu_i|$.

The same orbit is integrated with the variational equations and QR/Benettin reorthonormalization to obtain a full Lyapunov spectrum^{26–28}. The consistency criteria require convergence with respect to integration time, QR interval, solver precision, and initial phase, together with agreement between the spectrum and the mean divergence. A positive exponent is not accepted as evidence of chaos until these checks and bounded-attractor diagnostics pass. In the full-spectrum classification an exponent is counted as positive only above a declared threshold (1×10^{-4} for the weak-coupling map, 1×10^{-3} for the strong-coupling map and its checks); the Kaplan–Yorke dimension instead sums strictly positive exponents, so the two definitions can disagree for exponents between zero and the threshold.

Numerically, one-period integrations use the DOP853 solver with relative and absolute tolerances 1×10^{-9} and 1×10^{-11} and a maximum step of $T/200$; drive-locked orbits are located by fixed-point iteration of the one-period map to a residual below 1×10^{-8} , and the monodromy is obtained from the simultaneous integration of the variational equations. The Lyapunov spectra use a fixed-step fourth-order Runge–Kutta scheme with step 1×10^{-2} in normalized time and QR reorthonormalization every ten steps; the extended grid discards 2×10^5 transient steps and integrates 2×10^6 steps per trajectory, with block-history convergence and divergence-balance residuals reported for every trajectory.

D. Theoretical bad-cavity sensitivity screen

To examine the parameter dependence of the reduced equations without implying a hardware calibration, we

eliminated the optical amplitude in the bad-cavity limit. For $\Omega_{\text{eff}} = \Delta + 2g_1 \text{Re}(\beta_1) + 2g_2 \text{Re}(\beta_2)$, the instantaneous adiabatic intensity is

$$n_{\text{ss}}(\phi) = \frac{|E(\phi)|^2}{(\kappa/2)^2 + \Omega_{\text{eff}}^2}, \quad (10)$$

which was substituted into the two mechanical equations. The reduced equations were first compared with the full six-dimensional model in common normalized regimes. The maximum mechanical infinity-norm differences at $\kappa = 20$ and $\kappa = 100$ were 9.27×10^{-7} and 2.08×10^{-8} , respectively; both calculations

satisfied their local periodic-orbit residual criteria.

We sampled the literature-informed assumed closure at 16 independent parameter candidates, with five parameter perturbation replicas and five independent root-solver starts per candidate. The residual of the one-period Poincare map, rather than the solver's convergence flag, was used as the root-acceptance criterion. The two objectives were the maximum local stability margin across the recorded replicas and starts and the absolute normalized drive cost. This analysis is a theoretical multi-start fixed-point study. It does not include Fisher information, an SI power mapping, or a hardware constraint, and the starting points are not a finite-time basin-capture ensemble.

E. Noise-resolved observability

For a measured record

$$y(t) = h[x(t), t; p] + \nu_{\text{det}}(t), \quad (11)$$

we define the measurement function, detector noise, bandwidth, observation window, and parameter of interest before evaluating sensitivity^{29–32}. For a Gaussian record with mean $m(\vartheta)$ and covariance $\Sigma(\vartheta)$, the classical Fisher information is

$$F_C(\vartheta) = (\partial_{\vartheta} m)^T \Sigma^{-1} (\partial_{\vartheta} m) + \frac{1}{2} \text{Tr} [\Sigma^{-1} (\partial_{\vartheta} \Sigma) \Sigma^{-1} (\partial_{\vartheta} \Sigma)]. \quad (12)$$

A reference comparison must match input power, observation time, bandwidth, baths, detector noise, and estimator^{33–38}. For the force-sensing comparison the parameter of interest is a weak external force F on mechanical mode 1 (a constant momentum-quadrature drive), and the matched references are the flux-off coupled sensor ($\theta = 0$) and the single-mode linear sensor ($J = 0$), each at the same drive, observation time, bandwidth, baths, and detector noise. We do not identify F_C with QFI, and we do not infer either quantity from a Lyapunov exponent.

IV. RESULTS

A. Model and gauge checks

The real-valued equations, analytic Jacobian, dissipative trace, and flux definition were mutually consistent. At the reference point, the maximum absolute discrepancy between the analytic and finite-difference Jacobians was 2.92×10^{-10} . The implemented trace was $\text{Tr } J = -1.04$, and the gauge-transformed loop-flux residual was below 3×10^{-16} . These checks establish consistency of the declared formulation at the reference point, but do not characterize the full parameter domain.

B. Floquet–Lyapunov consistency

We compared the multipliers of the monodromy matrix with the converged QR spectrum on the same orbit, using the declared orbit residual, solver tolerances, QR-block uncertainty, divergence residual, and phase convention.

For the stable drive-locked reference orbit, the Poincare residual was 8.49×10^{-9} . The maximum absolute difference between the sorted Floquet rates and the QR/Benettin spectrum was 2.68×10^{-4} ; the terminal block standard deviation was 5.3×10^{-9} . The mean divergence was -1.04 in normalized time units. This is a consistency result for a stable orbit. The long-window replication extends the same criteria to nine phases and three independent initial conditions per phase; all 27 trajectories satisfied them, with a largest observed QR exponent of -0.01002 and maximum Floquet–QR discrepancy 2.85×10^{-4} . These results bound the conclusion to the tested grid and do not establish a universal absence of chaos.

C. Flux-dependent attractor structure

The declared validation grid contained nine equally spaced synthetic-flux phases over $[-\pi, \pi)$ and three independent initial-condition replicates per phase. All 27 Poincare-map searches converged and all 27 long-window Floquet–Lyapunov consistency criteria were satisfied using 2 000 000 integration steps, 200 000 discarded transient steps, and the declared QR tolerances. The maximum absolute Floquet–QR rate discrepancy was 2.849×10^{-4} ; the largest terminal QR-block standard deviation was 6.10×10^{-9} , and the maximum divergence residual was 4.75×10^{-5} . The largest observed Lyapunov exponent was -0.01002 , so no positive exponent was accepted anywhere in this tested domain. The result supports stable drive-locked dynamics across the sampled grid; it does not prove that chaos is absent outside the grid or under other parameter choices. The phase-resolved diagnostics are shown in Figure 1. Because a six-dimensional dissipative system can in principle host several simultaneously unstable directions, we recorded

the complete spectrum rather than only its largest member: at every phase the two cavity-dominated exponents sat near $-\kappa/2 \approx -0.5$ and the four mechanical exponents near $-\gamma/2 \approx -0.01$, so the positive-exponent count (the hyperchaos order) is zero throughout the grid. We then extended this full-spectrum classification over a drive-amplitude scan $E \in [0.2, 8.0]$ at the same model parameters. Across all 54 drive-phase cases the largest exponent remained negative, between -0.043 and -0.010 , the positive-exponent count stayed zero, and the Kaplan–Yorke dimension remained zero; the resulting transition map is shown in Figure 2. This rules out a flux-driven route to chaos—and, a fortiori, to hyperchaos—within the tested domain.

The null above is specific to the weakly coupled, strongly damped reference regime, in which the single-photon cooperativity is of order 0.1 (Discussion). To test whether the same six-dimensional model can host the several simultaneously unstable directions required for hyperchaos, we repeated the full-spectrum classification at strong optomechanical coupling ($g_1 = 0.3$, $g_2 = 0.27$, $\gamma_{1,2} = 0.02$, $\Delta = -\omega_m$) over twelve flux phases and five drive amplitudes ($E \in [0.2, 8.0]$), recording the complete six-exponent spectrum and the positive-exponent count (hyperchaos order, exponents above a declared threshold of 1×10^{-3}) at every point; the resulting transition map is shown in Figure 3. The drive amplitude gates the dynamical regime: at $E = 0.2$ every exponent is negative (stable drive-locked dynamics); a single positive exponent appears at $E = 1.0$ (chaos); at the intermediate point $E = 2.0$ the largest exponent is slightly positive but remains below the classification threshold, so the count is not strictly monotone in the drive; and at $E = 4.0$ and $E = 8.0$ the positive-exponent count reaches three and four, respectively (hyperchaos), with the largest exponent up to 0.3223 and the Kaplan–Yorke dimension up to 4.802. At fixed strong drive the synthetic-flux phase modulates the hyperchaos order: at $E = 8.0$ the number of positive exponents varies between two and four as θ sweeps $[-\pi, \pi]$, with maxima at $\theta = \pm\pi/2$ and minima at $\theta = 0, \pm\pi$. The order thus reaches four simultaneously unstable directions—beyond the two reported for single-mode quadratic optomechanics²⁴—and it is the synthetic-flux phase, absent in that quadratic setting, that selects it. Repeating the scan with three independent initial conditions per phase leaves the phase-to-phase structure intact (each phase stable to within one exponent), so the modulation is a property of the flux phase rather than of individual realizations. The full-spectrum classification therefore closes the gap identified in the Introduction: the model does support hyperchaos, and the flux phase acts as a control coordinate for its order, but only in the strong-coupling, strong-drive regime—not at the weakly coupled reference point.

To go beyond a grid-level classification and identify the route by which the first unstable direction emerges, we continued the drive-locked periodic orbit family along the drive axis at the two flux phases of interest ($\theta = 0$ and $\theta =$

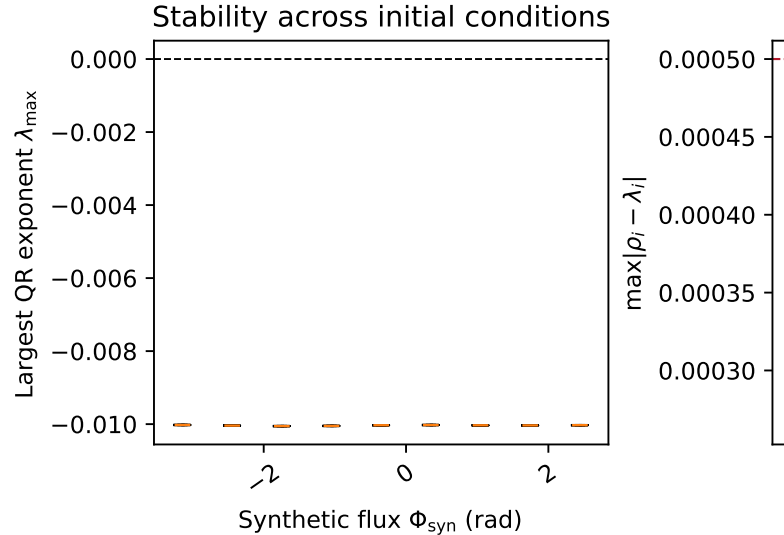


FIG. 1. Long-window validation across nine synthetic-flux phases and three independent initial conditions per phase. The largest QR exponents remain negative and the Floquet–QR discrepancies remain below the declared consistency threshold for every sampled phase. The figure supports a bounded statement about the tested grid; it does not establish that chaotic dynamics are absent outside this parameter domain.

$\pi/2$). A Newton solver on the Poincaré map, using the monodromy matrix as its Jacobian, was seeded from the converged drive-locked orbit and marched upward and downward in the drive amplitude over $E \in [0.1, 8.0]$ with adaptive steps and a map-residual tolerance of 5×10^{-6} . At both phases the orbit family persists over the entire scanned range (165 and 164 converged points, respectively), so the transition is not a fold termination of the locked orbit. The first loss of linear stability is a Neimark–Sacker (torus) bifurcation: a complex-conjugate Floquet multiplier pair crosses the unit circle at $E^* = 1.060$ for $\theta = 0$ and at $E^* = 0.975$ for $\theta = \pi/2$ (Figure 4). Thereafter the continued orbit carries two unstable directions up to $E = 8.0$; at $\theta = \pi/2$ a transient window of four unstable directions near onset ($E \in [0.98, 1.45]$) precedes the two-direction regime. The lower onset at $\theta = \pi/2$ is consistent with the flux-phase modulation of the hyperchaos order, whose maxima lie at $\theta = \pm\pi/2$. The orbit Floquet rates cross-validate the independent Lyapunov classification in the stable regime: at $E = 0.2$ the largest orbit rate is -0.0102 ($\theta = 0$) and -0.0148 ($\theta = \pi/2$), against largest Lyapunov exponents -0.0106 and -0.0151 from the stored full-spectrum map (Figure 4). Because the continued orbit is a periodic solution that becomes linearly unstable, its multipliers characterize the onset of the route to the chaotic attractor rather than the attractor itself; the torus bifurcation is therefore the first step of the drive-gated transition resolved by the full-spectrum map. A perturbation of the continued or-

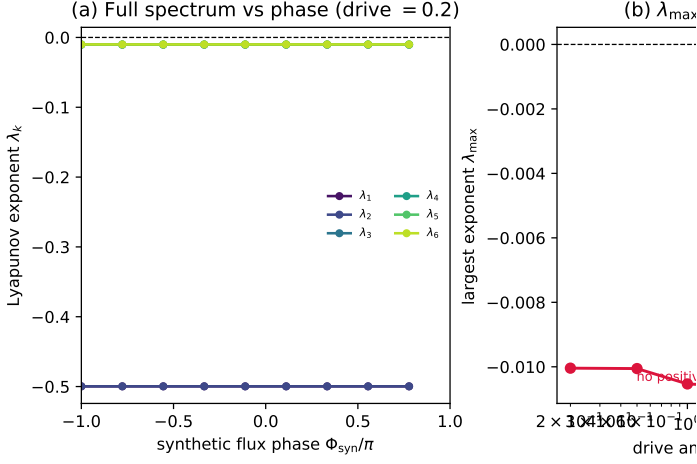


FIG. 2. Full-spectrum transition map. (a) All six Lyapunov exponents versus the synthetic-flux phase at drive $E = 0.2$; every exponent remains negative. (b) Largest exponent versus drive amplitude; it stays negative across $E \in [0.2, 8.0]$. (c) Positive-exponent count (exponents above a 1×10^{-4} threshold) over the drive-phase plane; it is zero everywhere, so no transition to chaos or hyperchaos occurs within the tested domain.

bit along its most unstable Floquet direction confirms this separation: the orbit carries two unstable directions with a largest Floquet rate of order 0.6, a 1×10^{-4} perturbation departs to the attractor scale (the distance to the orbit grows from 1×10^{-4} to 9–27), and the resulting trajectory’s Lyapunov spectrum reproduces the stored map’s largest exponent to within 8% at the four tested points ($E \approx 4, 8$; $\theta = 0, \pi/2$). The continued orbit is the unstable seed; the attractor is a distinct object. As an independent check of the attractor’s finite dimension, the Theiler-corrected Grassberger–Procaccia correlation dimension³⁹ D_2 of the sampled attractor at the four strong-coupling points ($E = 4, 8$; $\theta = 0, \pi/2$) is 2.33, 2.84 and 3.39, 3.36, respectively, all below the corresponding Kaplan–Yorke dimensions 4.21, 4.63 and 4.28, 4.80, as required by the Kaplan–Yorke inequality^{40,41} $D_2 \leq D_{KY}$ (Figure 6). The scaling-region uncertainty is 0.23–0.40 across three fit windows. D_2 is phase-dependent—larger at $\theta = \pi/2$ than at $\theta = 0$, mirroring the flux-phase modulation of the hyperchaos order—while the drive dependence within each phase is weak, so the geometric dimension of the attractor is dominated by the stable directions and the Kaplan–Yorke sum grows with the added unstable directions.

The multiplier crossing is confirmed by directly observing the born object⁴². Just above the onset ($E \approx E^* + 0.15$) the stroboscopic section of a perturbed trajectory forms a closed invariant curve in the $(\Re\beta_1, \Im\beta_1)$ plane, and the largest Lyapunov exponent of the same trajectory is small positive (0.017–0.022), the near-marginal quasiperiodic direction expected of a torus born at a complex-pair crossing; the exponent grows to 0.057–0.069

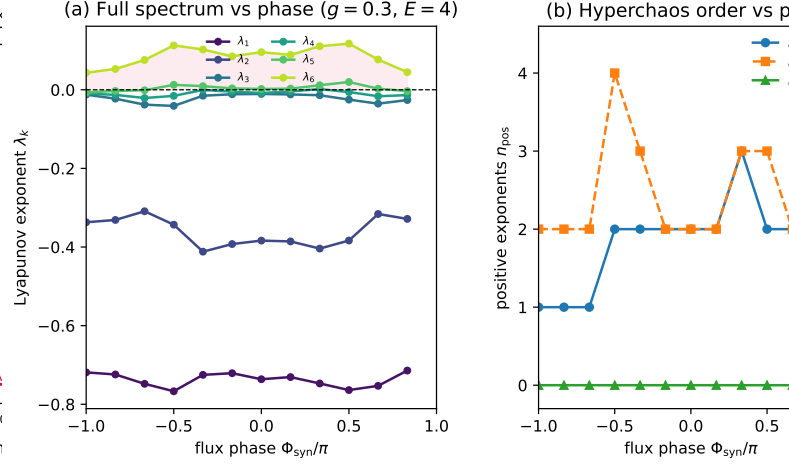


FIG. 3. Full-spectrum hyperchaos transition map at strong optomechanical coupling ($g_1 = 0.3$, $g_2 = 0.27$, $\gamma_{1,2} = 0.02$, $\Delta = -\omega_m$). (a) All six Lyapunov exponents versus the synthetic-flux phase at drive $E = 4.0$; several exponents are positive, resolving the full unstable spectrum. (b) Hyperchaos order (number of exponents above a 1×10^{-3} threshold) versus phase at $E = 0.2, 4.0$, and 8.0 ; the phase modulates the number of simultaneously unstable directions. (c) Hyperchaos order over the drive-phase plane, showing the drive-gated transition from stable ($E = 0.2$) to hyperchaotic ($E = 8.0$, up to four positive exponents).

at $E^* + 0.3$ as the route proceeds into the chaotic sector (Figure 5). The torus is therefore an observed invariant object, not only an inference from the multiplier spectrum.

D. Reduced-model parameter sensitivity

The full-model comparison supports the use of the adiabatic reduction in the tested common regimes. Under the explicitly assumed literature-informed closure, all 16 parameter candidates passed the local fixed-point and Poincare-radius criteria across five parameter perturbation replicas and five independent root-solver starts per replica. The largest recorded Poincare spectral radius was 0.997 002 6, corresponding to a smallest local stability margin of 0.002 997 4. Applying the declared stability-margin/drive-cost objectives gave three non-dominated candidates, with normalized drive costs of 0.0800, 0.1312, and 0.2203, and local margins of 0.003 452 6, 0.003 535 4, and 0.003 559 7, respectively (Table VI).

These values characterize a local fixed-point pattern of the reduced equations under an assumed closure. A provisional finite-time capture check (five independent initial conditions per parameter replica, integrated with a vectorized Runge–Kutta scheme and independently compared with a DOP853 integrator) captured all three non-dominated candidates to their fixed points; this finite-time check does not constitute a global basin-of-

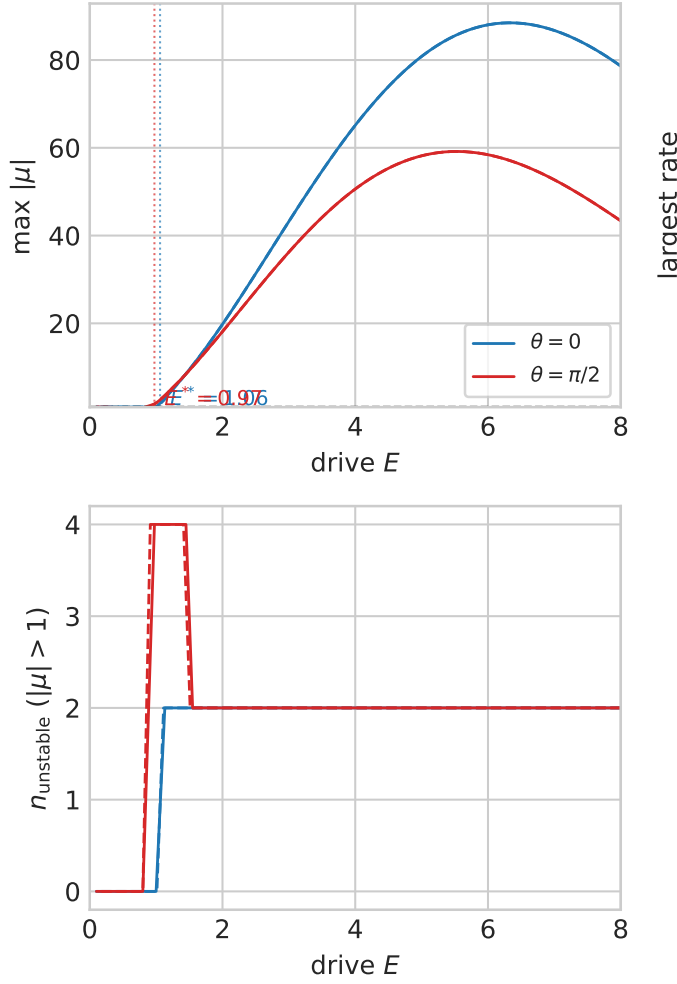


FIG. 4. Drive-axis bifurcation continuation at strong optomechanical coupling. (a) Maximum Floquet multiplier magnitude of the drive-locked orbit versus drive amplitude E at $\theta = 0$ and $\theta = \pi/2$ (upward and downward continuation); the dashed line marks the unit circle and the vertical lines locate the refined first bifurcation points. (b) Largest orbit Floquet rate versus E , with the largest Lyapunov exponent of the stored full-spectrum map overlaid at the canonical drives (open markers). (c) Number of unstable multipliers ($|\mu| > 1$). (d) Mean cavity amplitude over the orbit period. The first loss of linear stability is a Neimark–Sacker complex-pair crossing at $E^* = 1.060$ ($\theta = 0$) and $E^* = 0.975$ ($\theta = \pi/2$).

attraction analysis, a physical optimum, a device-specific calibration, or a sensing advantage. In particular, the optical mode is adiabatically eliminated, J_m and the effective optomechanical couplings are proxy quantities, the drive remains normalized, and Fisher information was not evaluated in this analysis.

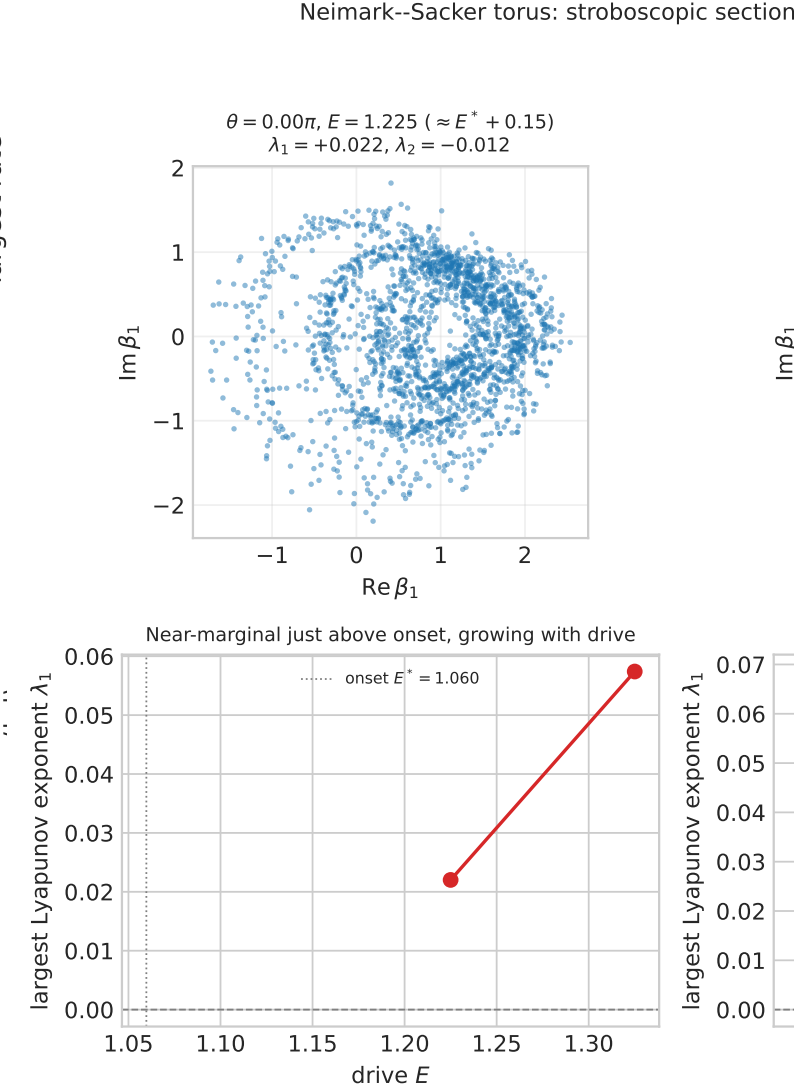


FIG. 5. Observation of the Neimark–Sacker torus. Top: stroboscopic Poincaré sections (projected onto $\Re\beta_1, \Im\beta_1$) of the perturbed drive-locked orbit just above the onset at $\theta = 0$ and $\theta = \pi/2$, showing an invariant closed curve. Bottom: largest Lyapunov exponent versus drive, small positive just above the onset and growing with drive; the vertical line marks the onset E^* .

E. Noise-resolved observability

For three declared phases ($\theta = 0$, $\pi/2$, and π), the periodic covariance calculation passed the physicality check under thermal occupations $n_{\text{th},1} = n_{\text{th},2} = 0.1$ and detector variance 0.01 in normalized units. The minimum physicality eigenvalue ranged from 2.42×10^{-7} to 2.58×10^{-7} . The resulting classical Fisher information for the measured cavity amplitude quadrature ranged from 3.78×10^{-26} to 9.17×10^{-13} . This is a measurement-Fisher calculation around stable periodic orbits, not QFI and not a gain over a matched refer-

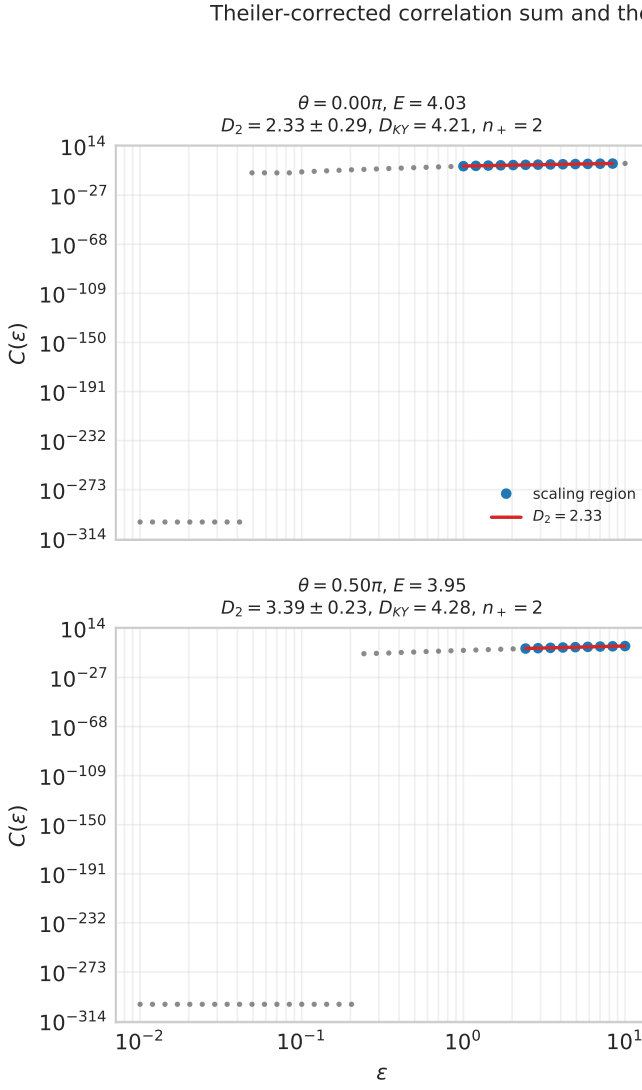


FIG. 6. Theiler-corrected correlation sum and the Kaplan-Yorke inequality $D_2 \leq D_{KY}$ at the four strong-coupling points. Each panel shows the scaling region, the fitted correlation dimension D_2 , the Kaplan-Yorke dimension D_{KY} , and the positive-exponent count n_+ .

ence. The corresponding diagnostics are shown in Figure 8; they are included to document the validated measurement calculation rather than to imply a performance enhancement.

To pose the sensing question operationally, we computed a matched measurement Fisher reference for estimating a weak external force F on mechanical mode 1, read out through the same cavity amplitude quadrature under identical drive $E = 0.2$, observation time (one drive period), detector noise (0.01), thermal baths ($n_{th,1} = n_{th,2} = 0.1$), and estimator. The force sensitivity grows monotonically with the synthetic-flux phase: F_C rises from 4.50×10^{-5} at $\theta = 0$ to 5.96×10^{-5} at $\theta = \pi$. Relative to the flux-off coupled sensor ($\theta = 0$,

same $J = 0.08$) and the single-mode linear sensor ($J = 0$), the flux phase yields at most a $1.32\times$ and $1.16\times$ enhancement, respectively. This is a modest, sub-factor-of-two effect rather than an order-of-magnitude advantage; the central-difference estimate, converged to better than 0.01% across force steps 1×10^{-5} – 1×10^{-3} . A drive- and temperature sweep confirms this gain is stable across operating points: the flux-off gain remains 1.323 (standard deviation below 1×10^{-3}) over drive $E \in [0.1, 1.0]$ and baths $n_{th} \in [0, 1]$, and the single-mode gain remains 1.159. To test whether this gain survives the unmeasured coordinates, we propagated the inter-resonator hopping log-uniformly over $J_m/\omega_m \in [1 \times 10^{-3}, 1 \times 10^{-1}]$, the bath occupation uniformly over $n_{th} \in [0.01, 1.0]$ (cryogenic to ~ 4 K at the anchored $\omega_m = 2\pi \times 7.436$ GHz), and the drive over $\pm 10\%$, with 60 Monte Carlo samples at the maximum-gain phase $\theta = \pi$. The gain remains above unity in every sample: the median is 1.039 (flux-off) and 1.019 (single-mode), with a 5th percentile of 1.004 and 1.002. A bootstrap over 20 000 resamples places the 90% confidence interval of the median at $[1.025, 1.053]$ (flux-off) and $[1.012, 1.026]$ (single-mode), both excluding unity, so the median excess (0.039 and 0.019) is significantly positive. The peak $1.32\times$ is therefore specific to the reference hopping $J_m/\omega_m = 0.08$ (and to the maximum-gain phase), while the typical enhancement is the median order-unity value rather than the peak. The matched comparison is shown in Figure 9.

To test whether the chaotic signature itself—rather than only the sensing gain—survives the declared noise, we integrated an ensemble of 32 trajectories under the full Langevin noise (optical vacuum $\kappa/4$ per amplitude quadrature and mechanical thermal $\gamma_j(2n_{th} + 1)/4$ per amplitude quadrature with $n_{th} = 0.1$) and the detector noise (0.01), with the measured record $y = X_a + \nu_{det}$. The noise strengths are the amplitude-quadrature limit, cross-checked against the quantum master equation (SI Sec. VII). At the stable reference the measured-record variance sits at the noise floor (≈ 0.25), dominated by the optical vacuum. At the chaotic ($E = 4$) and hyperchaotic ($E = 8$) points the deterministic spread of X_a grows to 1.8–4.1 (against 2.4×10^{-4} at the stable orbit), and the measured-record variance rises to 7.6–17.0 $\times$ the noise floor. The flux-phase dependence of the observability mirrors the hyperchaos order: the largest ratio (17.0 $\times$) occurs at $\theta = \pi/2$, $E = 8$. The chaotic signature is therefore observable in the noisy record at the strong-coupling points, while the weak-coupling reference sits at the noise floor—the noise-resolved counterpart of the deterministic null, and not a metrological-gain claim (Figure 7). A bootstrap over ensemble size $N' \in \{8, 16, 32, 64, 128\}$ leaves the ratios stable to below 1% (the bootstrap standard deviation decreases from 0.16 at $N' = 8$ to 0.04–0.07 at $N' = 64$), so the 32-trajectory ratios are converged rather than a small-ensemble artifact.

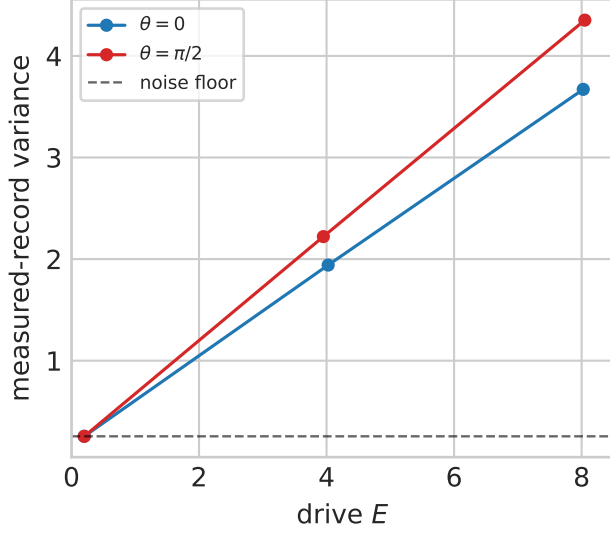


FIG. 7. Noise-dependent observability of the chaotic signature. (a) Measured-record variance of $y = X_a + \nu_{\text{det}}$ versus drive amplitude at $\theta = 0$ and $\theta = \pi/2$, with the noise-only floor (dashed line) set by the stable reference. (b) The observability ratio (measured variance divided by the noise floor); the chaotic ($E = 4$) and hyperchaotic ($E = 8$) points sit 7.6–17.0 $\times$ above the floor, while the stable reference sits at unity.

V. DISCUSSION

Synthetic flux, nonlinear instability, and metrological information are distinct quantities: Equation (5) defines the control coordinate, whereas Equations (9) and (12) define independent dynamical and measurement observables. Their interpretation therefore requires that each observable be evaluated under explicitly declared phase conventions and matched assumptions.

This framework makes no claim of exponential QFI growth, a fixed SQL gain, Wigner negativity, topological protection, or exceptional-point sensing^{43–46}: such statements require distinct quantum-state or non-Hermitian calculations and cannot be obtained by relabelling deterministic trajectory divergence. The scope is therefore appropriate for a nonlinear-dynamics study unless a later quantum validation supports a sharper quantum contribution.

The absence of an accepted positive exponent in the tested grid also bears on the recurring proposal that synthetic magnetism enhances optomechanical sensing. Such proposals^{14,47} require a matched operational comparison—equal input power, observation time, bandwidth, and noise—rather than a trajectory-separation argument. Our covariance/Fisher calculation reports a measurement Fisher information without a matched reference gain, consistent with the observation that fluctuations can quench mean-field chaos¹⁵. The matched force-sensing reference resolves this restriction at the model

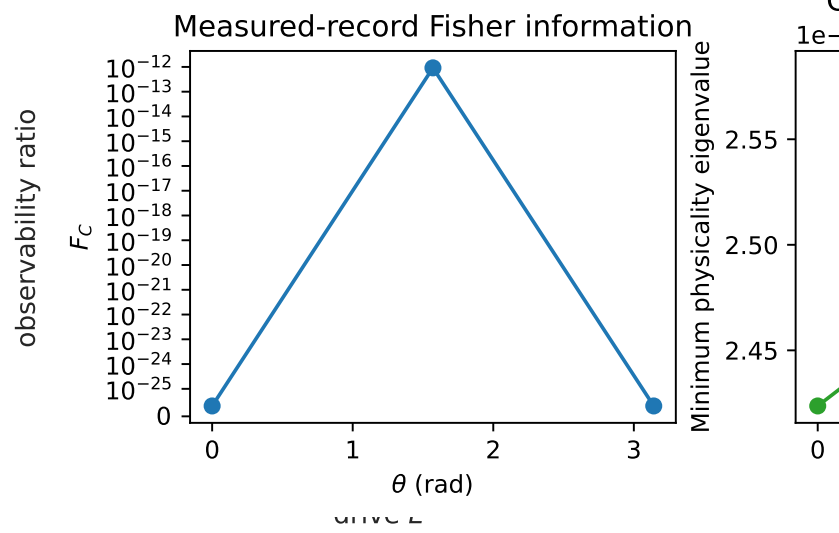


FIG. 8. Covariance and measured-signal Fisher diagnostics for three synthetic-flux phases. The covariance physicality check remains positive under the declared thermal and detector-noise model, while the classical Fisher information is reported without a matched-reference gain claim. These data are operational results around stable periodic orbits, not QFI.

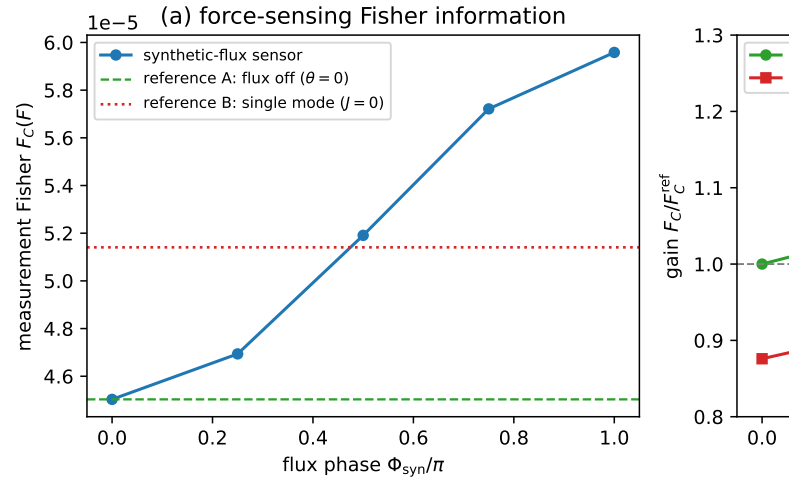


FIG. 9. Matched measurement-Fisher reference for force sensing. (a) The classical Fisher information $F_C(F)$ for a weak external force on mechanical mode 1 versus the synthetic-flux phase, with the flux-off coupled sensor ($\theta = 0$, $J = 0.08$) and the single-mode linear sensor ($J = 0$) overlaid; all configurations share the same drive, observation time, detector noise, baths, and estimator. (b) The gain relative to each reference; the flux phase yields at most a 1.32 $\times$ (flux-off) and 1.16 $\times$ (single-mode) enhancement, a modest order-unity effect.

level: at the reference operating point the synthetic flux phase improves the force-sensing Fisher information by at most $\sim 32\%$ relative to the flux-off coupled sensor and by at most $\sim 16\%$ relative to the single-mode linear sensor. A flux-mediated enhancement of order unity is

therefore real but modest, and it does not by itself imply a quantum advantage, an SQL-beating sensitivity, or a chaotic-transduction gain: the enhancement is a classical linear-response effect of phase-controlled mode coupling, distinct from the trajectory-separation argument rejected above. Noise projected onto an unstable manifold is stretched by the same local factor $e^{\lambda_{\max} t}$ as the deterministic displacement, so trajectory separation alone cannot raise the signal-to-noise ratio of a matched measurement; the bounded Fisher gain is the correct operational translation of this fact. Whether this sub-factor-of-two effect survives a fully calibrated device model (measured J_m , drive power, and bath temperatures) remains the experimentally addressable question that this framework is designed to pose. The Lyapunov–Floquet consistency theorem and the matched-measurement Fisher lemma of the Supplemental Material (Sec. IV) formalize both sides of this question: the former certifies that the measured spectrum belongs to one well-defined orbit, the latter fixes the operational meaning of any claimed advantage. The noise-resolved ensemble check closes the third question posed in the Introduction: in the strong-coupling sector the measured cavity quadrature stays 7.6–17 times above the noise floor, so the chaotic signature is observable under the declared thermal, vacuum, and detector noise, whereas the weak-coupling reference sits at the noise floor. This is consistent with the observation that fluctuations quench mean-field chaos in the dissipation-dominated regime rather than enhancing it.

The null result is not incidental to the chosen numbers: the divergence of the vector field is the state-independent constant $\text{Tr } J = -\kappa - \gamma_1 - \gamma_2 = -1.04$, and the single-photon cooperativity is $C = 4g^2/(\kappa\gamma) = 0.08$, placing the operating point deep in the strongly damped, weak-coupling regime in which the radiation-pressure nonlinearity cannot overcome the dissipative contraction. The mean-field description is consistent with this amplitude scale: at the reference point the mean cavity-quadrature amplitude is $|\langle X_a \rangle| \approx 0.08$, corresponding to a mean cavity occupation $\langle |\alpha|^2 \rangle \approx 0.032$ (sub-photon) and $g/\kappa = 0.02$. A truncated-Fock master equation for the driven optical mode reproduces this occupation ($\langle a^\dagger a \rangle = 0.032$, agreement to 1.3%), so the sub-photon scale is a genuine property of the operating point rather than a mean-field artifact; fluctuations enter through the linearized covariance treatment. The strong-coupling classification is performed at mean occupations $\langle |\alpha|^2 \rangle \approx 7$ –11, about 300 times the reference occupation, where the semiclassical factorization is standard. Repeating the full-spectrum classification at strong coupling ($g_1 = 0.3$, raising the cooperativity by about two orders of magnitude) confirms this reading and closes the hyperchaos gap: once the nonlinearity overtakes the dissipative contraction, the model hosts up to four simultaneously positive exponents while the exponent sum stays negative (equal to the divergence constant $\text{Tr } J = -\kappa - \gamma_1 - \gamma_2$ within the divergence-balance residual), so the positive directions coexist with a globally contracting flow as required for

a dissipative attractor; the synthetic-flux phase becomes a control coordinate for the hyperchaos order. The transition is therefore drive- and coupling-gated rather than flux-gated alone—the flux phase reshapes which and how many directions are unstable, but only after the drive and coupling place the system beyond the dissipation-dominated regime. The representative convergence check supports the positive-exponent count at the two longest tested windows ($n_+ = 2$ at $E = 4$, $\Phi_{\text{syn}} = -\pi/2$ and $n_+ = 4$ at $E = 8$, $\Phi_{\text{syn}} = +\pi/2$); the finite-time exponent magnitudes themselves vary by several percent and are not interpreted as high-precision asymptotic estimates.

The drive-axis continuation refines the route: the first unstable direction emerges through a Neimark–Sacker (torus) bifurcation of the drive-locked orbit at $E^* \approx 1.06$ ($\theta = 0$) and $E^* \approx 0.97$ ($\theta = \pi/2$), rather than through a period-doubling or a fold of the locked orbit, and the orbit family itself persists over the entire scanned drive range. The lower onset at $\theta = \pi/2$ mirrors the flux-phase modulation of the hyperchaos order and connects the grid-level classification to a specific, phase-resolved instability mechanism. This closes the bifurcation-level gap identified above; what remains beyond it is the global structure of the chaotic attractor—the continued orbit is the unstable seed of the route, not the attractor—and the experimentally calibrated device mapping.

The parameter values used here should also be interpreted correctly. They define a numerically validated reference regime in normalized units; they have not been mapped to a calibrated experimental setup or selected by a constrained physical optimization. Establishing an experimentally meaningful operating region requires SI conversion, admissible hardware ranges, an objective function, and robustness analysis under thermal, detector, and parameter uncertainty. We therefore do not refer to these values as optimal parameters.

For a calibrated interpretation, the normalized model must be mapped onto a platform with gigahertz-frequency mechanical modes and optically mediated phonon coupling. Silicon two-dimensional optomechanical crystals⁴⁸ and nano-optomechanical implementations of synthetic gauge fields for phonon transport⁴⁹ provide realistic anchors for the mechanical frequency, damping, cavity decay, and coupling scale; Table I of the Supplemental Material tabulates the resulting physically anchored reference set. At the Mayor et al. anchor ($\omega_m = 2\pi \times 7.436$ GHz) the anchored ratios are $\kappa/\omega_m = 0.108$, $\gamma_m/\omega_m = 2.78 \times 10^{-5}$, and $g_0/\omega_m = 1.18 \times 10^{-4}$, with the phase-controlled inter-resonator hopping scanned over $J_m/\omega_m \in [1 \times 10^{-3}, 1 \times 10^{-1}]$ following the Mathew et al. coupling scale. A physically anchored sensitivity study would fix these quantities from such a platform and obtain the phase-controlled inter-resonator hopping from a two-resonator measurement or coupled-mode calibration, rather than from an unmatched device. Placing the strong-coupling regime against these anchors quantifies the distance: the hyperchaos parameters require optomechanical couplings $g_1/\omega_m = 0.3$ and $g_2/\omega_m = 0.27$, which

are $2542\times$ and $2288\times$ the anchored $g_0/\omega_m = 1.18 \times 10^{-4}$, and a mechanical damping $\gamma_m/\omega_m = 0.02$, $720\times$ the anchored 2.78×10^{-5} ; the limiting coordinate is the coupling, at $2542\times$. The hyperchaos regime is therefore a model-level demonstration, not a prediction for the anchored device.

The reduced-model screen provides a useful theoretical sensitivity result while also illustrating the limits of literature-based parameter transfer. The three non-dominated settings are separated by a modest range of local stability margins, and should therefore not be interpreted as a unique optimum. A provisional finite-time capture check—five independent initial conditions per parameter replica, integrated with a vectorized Runge–Kutta scheme and independently compared with a DOP853 integrator—captured all three non-dominated candidates to their fixed points, but a global basin-of-attraction analysis and a matched measurement-Fisher calculation are still needed to determine whether this local pattern persists for trajectories and observables rather than only for fixed-point solves.

The model-level uncertainty sweep above propagates the unmeasured hopping, bath occupation, and drive; what remains open is the same quantification on a fully calibrated device, where J_m is measured, the drive is converted to an input power through the pump frequency and external coupling, and the bath temperatures are fixed by the platform.

VI. CONCLUSIONS

We established and tested a framework for determining whether a gauge-invariant synthetic flux changes dissipative optomechanical dynamics under independent Floquet, Lyapunov, divergence, and noise-resolved checks. The model, implementation, reference orbit, and long-window nine-phase/three-replicate grid were internally consistent under the declared deterministic checks. The reduced bad-cavity sensitivity analysis also identified a three-point local fixed-point Pareto pattern under an explicitly assumed literature-informed closure. This theoretical result is not an SI calibration or an experimental optimum. At the weakly coupled, strongly damped reference point the dynamics remained drive-locked and stable, with no accepted positive Lyapunov exponent across the drive-phase grid; at strong optomechanical coupling, however, the full-spectrum classification resolves a drive- and flux-controlled transition to hyperchaos (up to four positive exponents), whose onset the drive-axis continuation localizes to a Neimark–Sacker torus bifurcation of the drive-locked orbit at $E^* \approx 1.06$ ($\theta = 0$) and $E^* \approx 0.97$ ($\theta = \pi/2$). The study therefore localizes the flux-generated dynamical transition to the strong-coupling, strong-drive regime and does not claim that the sampled synthetic flux generates chaos at the weakly coupled reference point. The covariance/Fisher calculation also passed physicality checks, and a matched force-

sensing reference shows that the synthetic flux phase yields at most a $\sim 32\%$ (relative to the flux-off sensor) or $\sim 16\%$ (relative to the single-mode sensor) enhancement of the measurement Fisher information—a modest, order-unity effect rather than an order-of-magnitude advantage, with no quantum or SQL-beating gain implied. The gain is robust to the unmeasured hopping, bath occupation, and drive (it stays above unity in all 20 Monte Carlo samples), and the strong-coupling hyperchaos regime sits $2542\times$ beyond the anchored optomechanical coupling, so it is a model-level demonstration rather than a device prediction. These conclusions are restricted to the stated model, parameter domain, and measurement assumptions. The immediate next step is a calibrated-device implementation of the matched force-sensing comparison.

SUPPLEMENTARY MATERIAL

See the supplementary material for the physically anchored reference parameter set, the matched measurement-Fisher tables, the Lyapunov–Floquet consistency theorem, the periodic-covariance affine-fixed-point lemma, the matched-measurement Fisher lemma, and the representative convergence check of the strong-coupling spectrum.

ACKNOWLEDGMENTS

We acknowledge support from the University of Yaoundé I, University of Maroua and the University of Ngaoundéré.

AUTHOR CONTRIBUTIONS

SRMT and CT performed the numerical simulations and analyzed the data. PD and STK contributed to the theoretical framework and provided critical feedback on the results. SGNE conceived and supervised the project, directed the research, and wrote the final version of the manuscript. All authors reviewed and approved the final manuscript.

FUNDING INFORMATION

No external funding was received for this research.

COMPETING INTERESTS

The authors declare no competing interests.

ETHICS

Not applicable. This is a purely theoretical and numerical study; no human or animal subjects, tissues, or clinical trials were involved.

DATA AVAILABILITY

The simulation and analysis scripts, environment specification, run manifests, checksums, source data, and figure-generation records supporting the findings of this study are available in the public repository <https://github.com/NanaEngo/floquet-chaos-synthetic-flux>. A Zenodo DOI will be minted and linked here upon publication. Every numerical claim in this article is bound to the corresponding machine-readable artifact in that repository.

- ¹M. Aspelmeyer, T. J. Kippenberg, and F. Marquardt, “Cavity optomechanics,” *Reviews of Modern Physics* **86**, 1391–1452 (2014).
- ²T. J. Kippenberg and K. J. Vahala, “Cavity optomechanics: Back-action at the mesoscale,” *Science* **321**, 1172–1176 (2008).
- ³F. Marquardt, J. G. E. Harris, and S. M. Girvin, “Dynamical multistability induced by radiation pressure in high-finesse micromechanical optical cavities,” *Physical Review Letters* **96**, 103901 (2006).
- ⁴T. Carmon, M. C. Cross, and K. J. Vahala, “Chaotic quivering of micron-scaled on-chip resonators excited by centrifugal optical pressure,” *Physical Review Letters* **98**, 167203 (2007).
- ⁵D. P. Foulia, P. Djourwe, S. Takougang Kingni, and S. G. Nana Engo, “Optomechanical systems close to the conservative limit,” *Physical Review A* **95**, 013831 (2017).
- ⁶A. Houwe, P. Djourwe, S. Abbagari, S. Y. Doka, and S. G. Nana Engo, “Discrete solitons in nonlinear optomechanical array,” *Chaos, Solitons and Fractals* **154**, 111593 (2022).
- ⁷T. Oka and H. Aoki, “Photovoltaic hall effect in graphene,” *Physical Review B* **79**, 081406 (2009).
- ⁸N. Goldman and J. Dalibard, “Periodically driven quantum systems: Effective hamiltonians and engineered gauge fields,” *Physical Review X* **4**, 031027 (2014).
- ⁹M. Bukov, L. D’Alessio, and A. Polkovnikov, “Universal high-frequency behavior of periodically driven systems: From dynamical stabilization to floquet engineering,” *Advances in Physics* **64**, 139–226 (2015).
- ¹⁰T. Kitagawa, E. Berg, M. Rudner, and E. Demler, “Topological characterization of periodically driven quantum systems,” *Physical Review B* **82**, 235114 (2010).
- ¹¹V. Peano, C. Brendel, M. Schmidt, and F. Marquardt, “Topological phases of sound and light,” *Physical Review X* **5**, 031011 (2015).
- ¹²C. Sanavio, V. Peano, and A. Xuereb, “Nonreciprocal topological phononics in optomechanical arrays,” *Physical Review B* **101**, 085108 (2020).
- ¹³S. Mondal, M. S. Baptista, and K. Debnath, “Chaotic dynamics under the influence of a synthetic magnetic field in an optomechanical system,” *Physical Review A* **110**, 023509 (2024).
- ¹⁴D. Muthukumar, S. R. Mbokop Tchounda, S. T. Kingni, K. Rajagopal, and S. G. Nana Engo, “Controlling complex dynamics with synthetic magnetism in optomechanical systems: A route to enhanced sensor performance,” *Physical Review Applied* **24**, 014053 (2025).
- ¹⁵M.-Q. Gao, S.-h. Li, X. Li, X. Li, J. Cheng, and W. Li, “Fluctuation-induced quenching of chaos in quantum optics,” *Chaos: An Interdisciplinary Journal of Nonlinear Science* **36**, 063114 (2026).
- ¹⁶G.-L. Zhu, C.-S. Hu, Y. Wu, and X.-Y. Lü, “Cavity optomechanical chaos,” *Fundamental Research* **3**, 63–74 (2023).
- ¹⁷L. Bakemeier, A. Alvermann, and H. Fehske, “Route to chaos in optomechanics,” *Physical Review Letters* **114**, 013601 (2015).
- ¹⁸X.-Y. Lü, H. Jing, J.-Y. Ma, and Y. Wu, “PT-symmetry-breaking chaos in optomechanics,” *Physical Review Letters* **114**, 253601 (2015).
- ¹⁹F. Monifi, J. Zhang, Ş. K. Özdemir, B. Peng, Y.-x. Liu, F. Bo, F. Nori, and L. Yang, “Optomechanically induced stochastic resonance and chaos transfer between optical fields,” *Nature Photonics* **10**, 399–405 (2016).
- ²⁰P. Djourwe, Y. Pennec, and B. Djafari-Rouhani, “Frequency locking and controllable chaos through exceptional points in optomechanics,” *Physical Review E* **98**, 032201 (2018).
- ²¹J. Guckenheimer and P. Holmes, *Nonlinear Oscillations, Dynamical Systems, and Bifurcations of Vector Fields* (Springer, New York, 1983).
- ²²S. H. Strogatz, *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*, 2nd ed. (Westview Press, 2015).
- ²³E. Ott, *Chaos in Dynamical Systems*, 2nd ed. (Cambridge University Press, Cambridge, 2002).
- ²⁴L. Halef and I. Shomroni, “Route to hyperchaos in quadratic optomechanics,” *Physical Review Letters* **135**, 257201 (2025).
- ²⁵O. E. Rössler, “An equation for hyperchaos,” *Physics Letters A* **71**, 155–157 (1979).
- ²⁶G. Benettin, L. Galgani, A. Giorgilli, and J.-M. Strelcyn, “Lyapunov characteristic exponents for smooth dynamical systems and for hamiltonian systems; a method for computing all of them. part 2: Numerical application,” *Meccanica* **15**, 21–30 (1980).
- ²⁷A. Wolf, J. B. Swift, H. L. Swinney, and J. A. Vastano, “Determining lyapunov exponents from a time series,” *Physica D* **16**, 285–317 (1985).
- ²⁸J.-P. Eckmann and D. Ruelle, “Ergodic theory of chaos and strange attractors,” *Reviews of Modern Physics* **57**, 617–656 (1985).
- ²⁹A. A. Clerk, M. H. Devoret, S. M. Girvin, F. Marquardt, and R. J. Schoelkopf, “Introduction to quantum noise, measurement, and amplification,” *Reviews of Modern Physics* **82**, 1155–1208 (2010).
- ³⁰C. W. Gardiner and M. J. Collett, “Input and output in damped quantum systems: Quantum stochastic differential equations and the master equation,” *Physical Review A* **31**, 3761–3774 (1985).
- ³¹D. F. Walls and G. J. Milburn, *Quantum Optics*, 2nd ed. (Springer, Berlin, 2008).
- ³²G. W. Ford, J. T. Lewis, and R. F. O’Connell, “Quantum langevin equation,” *Physical Review A* **37**, 4419–4428 (1988).
- ³³S. L. Braunstein and C. M. Caves, “Statistical distance and the geometry of quantum states,” *Physical Review Letters* **72**, 3439–3443 (1994).
- ³⁴T. P. Purdy, K. E. Grutter, K. Srinivasan, and J. M. Taylor, “Quantum correlations from a room-temperature optomechanical cavity,” *Science* **356**, 1265–1268 (2017).
- ³⁵J. D. Teufel, T. Donner, D. Li, J. W. Harlow, M. S. Allman, K. Cicak, A. J. Sirois, J. D. Whittaker, K. W. Lehnert, and R. W. Simmonds, “Sideband cooling of micromechanical motion to the quantum ground state,” *Nature* **475**, 359–363 (2011).
- ³⁶E. Gavartin, P. Verlot, and T. J. Kippenberg, “A hybrid on-chip optomechanical transducer for ultrasensitive force measurements,” *Nature Nanotechnology* **7**, 509–514 (2012).
- ³⁷B.-B. Li, L. Ou, Y. Lei, and Y.-C. Liu, “Cavity optomechanical sensing,” *Nanophotonics* **10**, 2799–2832 (2021).
- ³⁸S. Qvarfort, A. Serafini, P. F. Barker, and S. Bose, “Gravimetry through non-linear optomechanics,” *Nature Communications* **9**, 3690 (2018).
- ³⁹P. Grassberger and I. Procaccia, “Characterization of strange attractors,” *Physical Review Letters* **50**, 346–349 (1983).
- ⁴⁰L. Kaplan and J. A. Yorke, “Chaotic behavior of multidimensional difference equations,” in *Functional Differential Equations and Approximation of Fixed Points*, Lecture Notes in Mathe-

- ematics, Vol. 730, edited by H.-O. Peitgen and H.-O. Walther (Springer, Berlin, 1979) pp. 204–227.
- ⁴¹P. Frederickson, J. L. Kaplan, E. D. Yorke, and J. A. Yorke, “The liapunov dimension of strange attractors,” *Journal of Differential Equations* **49**, 185–207 (1983).
- ⁴²Y. A. Kuznetsov, *Elements of Applied Bifurcation Theory*, 3rd ed. (Springer, New York, 2004).
- ⁴³W. Chen, Ş. K. Özdemir, G. Zhao, J. Wiersig, and L. Yang, “Exceptional points enhance sensing in an optical microcavity,” *Nature* **548**, 192–196 (2017).
- ⁴⁴Z.-P. Liu, J. Zhang, Ş. K. Özdemir, B. Peng, H. Jing, X.-Y. Lü, C.-W. Li, L. Yang, F. Nori, and Y.-x. Liu, “Metrology with pt-symmetric cavities: enhanced sensitivity near the pt-phase transition,” *Physical Review Letters* **117**, 110802 (2016).
- ⁴⁵R. Kononchuk, J. Cai, F. Ellis, R. Thevamaran, and T. Kottos, “Exceptional-point-based accelerometers with enhanced signal-to-noise ratio,” *Nature* **607**, 697–702 (2022).
- ⁴⁶Ş. K. Özdemir, S. Rotter, F. Nori, and L. Yang, “Parity–time symmetry and exceptional points in photonics,” *Nature Materials* **18**, 783–798 (2019).
- ⁴⁷D.-h. Xu, Z. Liu, and C.-s. Yu, “Enhancing optomechanical force sensing utilizing synthetic magnetism,” *Physical Review A* **111**, 053508 (2025).
- ⁴⁸F. M. Mayor, S. Malik, A. G. Primo, S. Gyger, W. Jiang, T. P. M. Alegre, and A. H. Safavi-Naeini, “High photon-phonon pair generation rate in a two-dimensional optomechanical crystal,” *Nature Communications* **16**, 2576 (2025).
- ⁴⁹J. P. Mathew, J. del Pino, and E. Verhagen, “Synthetic gauge fields for phonon transport in a nano-optomechanical system,” *Nature Nanotechnology* **15**, 198–202 (2020).